TT-E-522A
January 28, 1974
SUPERSEDING
Fed. Spec. TT-E-522
May 28, 1963

FEDERAL SPECIFICATION

ENAMEL, PHENOLIC, OUTSIDE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an outside phenolic enamel.

1.2 Classification. The enamel shall be of the following types as specified (see 6.2):

Type I - Lusterless
Type II - Semi-gloss

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking Of.
TT-T-775 - Tung Oil, Raw (China-Wood) (For Use in Organic Coating).
TT-V-121 - Varnish, Spar, Water-Resisting.

FSC 8010

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Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers,at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-T-704 - Treatment and Painting of Materiel.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Air Pollution Control District

Rule 66.

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, CA 90001.)

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American Society for Testing and Materials (ASTM) Standards:

B117	- Salt Spray (Fog) Testing.
E 28	- Softening Point by Ring-and-Ball Apparatus.
G 26	- Operating Light- and Water-Exposure Apparatus (Xenon-Arc Type) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Manufacturing Chemists' Association, Inc.

Manual L-1, Guide to the Precautionary Labeling of Hazardous Chemicals

(Application for copies should be addressed to the Manufacturing Chemists' Association, Inc., 1825 Connecticut Avenue, NW, Washington, DC 20003.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification. The solvents used in the formulation shall conform to the requirements of Rule 66.

3.1.1 Pigment. The choice of pigment is optional provided the end product conforms to this specification.

3.1.2 Raw tung oil. Raw tung oil shall conform to TT-T-775.

3.1.3 Phenolformaldehyde resin. The phenolformaldehyde resin shall conform to the requirements of table I.

Table I. Phenolformaldehyde resin

Characteristic	Requirement	
	Minimum	Maximum
Specific gravity at 25° C.	1.0	1.1
Softening point (ball and ring), °F.....	250	280
Ash, percent by weight....	---	0.2
Volatile matter at 293° C., percent by weight...	---	5
Color (using 50 percent solution in toluol) Gardner.....	---	6

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3.2 Manufacture. The component raw materials shall be mixed so as to produce a homogeneous product. The varnish portion of the enamel shall be made by processing 100 percent phenolformaldehyde resin with raw tung oil and thinners (conforming to the requirements of Rule 66) in the proportion of 100 pounds of resin to 25 gallons of raw tung oil. The resulting varnish shall be clear, stable, and transparent and shall conform to the requirements of table II.

Table II. Varnish

Characteristic	Requirement	
	Minimum	Maximum
Nonvolatile, percent by weight	53	55
Ash, percent by weight	---	0.5
Viscosity, Gardner-Holdt	E	G
Color, Gardner	---	17
Weight per gallon, pounds	7.2	7.4

3.3 Quantitative requirements. The enamel shall conform to the quantitative requirements of table III.

Table III. Quantitative requirements

Characteristic	Requirements	
	Minimum	Maximum
Nonvolatile content (percent by weight of enamel)	50	---
Fineness of grind	4	---
Water (percent by weight of enamel)	---	1.0
Rosin and rosin derivatives	---	0
60 degree specular gloss:		
Type I	---	8
Type II	25	40
Consistency, Krebs-Stormer:		
Shearing rate 200 rpm		
Grams	150	225
Equivalent, Krebs Units	75	85

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3.4 Qualitative properties.

3.4.1 Condition in container. The enamel in a freshly opened container shall be free from grit, skins, lumps, abnormal thickening, or livering and shall show no settling that cannot be redispersed with a paddle to a homogeneous state.

3.4.2 Storage stability.

3.4.2.1 Full container. A full closed container shall withstand 6 months' storage at any temperature from 0°F. to 105°F. without showing any livering, curdling, excessive bodying or settling that cannot be redispersed with a paddle to a homogeneous state. The enamel shall also show no loss in drying properties or any change in viscosity that could affect the working properties.

3.4.2.2 Partially filled container. A partially filled, closed container after 24 hours storage at room temperature shall show no skinning. A partially filled, closed container after 30 days storage at room temperature shall mix manually to a smooth, homogeneous state.

3.4.3 Working properties. When thinned in accordance with the suppliers recommendations, the enamel shall be suitable for brushing, dipping, or spraying and shall have good leveling properties.

3.4.4 Color. Color shall be as specified (see 6.2) and shall visually match the color numbers specified in table IV. White enamel shall have a directional reflectance greater than 83 percent.

3.4.5 Drying properties. When applied to steel panels and air-dried at room temperature, the enamel shall dry dust free in not more than 30 minutes, dry through in no more than 3 hours, and dry hard in not more than 16 hours.

3.4.6 Self-lifting properties. Recoating a steel or wood panel after the first coat has air-dried for 16 hours shall produce no film irregularity.

3.4.7 Baking properties. When tested as specified in 4.4.2, the steel panel, recoated after the first coat has air-dried for 16 hours, shall produce no film irregularity.

3.4.8 Immersion resistance. When applied to steel and wood panels and air-dried, the film shall show no blistering or loss of adhesion after immersion in distilled water for 21 days followed by 24 hours air-drying.

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3.4.9 Salt-spray resistance. When applied to steel and wood panels and air-dried, the film shall show no softening, blistering, peeling, embrittlement, or (in the case of steel panels) corrosion extending more than 1/8 inch beyond the score line after 300 hours exposure to salt spray.

3.4.10 Accelerated weathering. When applied to steel panels and air-dried, the film shall show no more than slight fading or chalking and no cracking after 300 hours exposure to accelerated weathering. The panel shall show no corrosion. When chalking occurs, the chalking shall be removed by gentle rubbing with a damp rag. Slight changes in the value or chroma of the color is permissible after chalk has been removed, but the hue shall remain constant, i.e., a blue-green may become lighter or darker or become more or less gray, but shall not become a yellow-green.

3.4.11 Flexibility. When applied to a steel panel and baked, the enamel shall not crack or peel when bent 180 degrees over a 1/4 inch mandrel.

3.4.12 Contrast ratio (dry opacity). When tested as specified in 4.4.1 the minimum contrast ratio for the enamel shall be as specified in table IV.

Table IV. Contrast ratio of selected colors of enamel

Color	Color No. FED. STD. No. 595	Dry film thickness in mils	Contrast ratio minimum
Brown	20117	1	0.98
Brown	30118	1	0.98
Brown	30099	1	0.98
Brown	30111	1	0.98
Red	31136	1.5	0.95
Yellow	33538	1.5	0.95
Green	34087	1	0.98
Green	34108	1	0.96
Blue	35044	1	0.96
Blue	35042	1	0.98
Blue	34158	1	0.98
Gray	36231	1	0.98
Gray	36118	1	0.98
Black	37038	1	0.98
White	37875	1.5	0.92

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling and inspection. Sampling and inspection shall be in accordance with FED. TEST STD. No. 141, methods 1021 and 1031.2.

4.3.2 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.1 through 4.4.6. Failure of any test shall be cause for rejection of the lot from which the sample was taken.

4.3.3 Test conditions. Except as otherwise specified herein, physical properties of the enamel shall be determined on packaged material which has been reduced for spraying with a thinner conforming to the requirements of Rule 66.

4.4 Test procedures.

4.4.1 Qualitative. The following tests shall be conducted in accordance with FED. TEST METHOD STD. No. 141, methods as specified in table V. Failure of any test result to fall within the ranges specified in 3.1.3, 3.2 or 3.3 shall constitute failure of the applicable test.

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Table V. Tests and methods

Examination or test	Applicable method in FED. STD. No. 141	Requirements paragraphs
Contrast ratio	4121	3.4.12
Weight per gallon of varnish	4184	table II
Viscosity, Gardner-Holdt	4271	table II
Moisture content	4081	table III
Color of varnish, Gardner	4248	table II
Condition in container	3011	3.4.1
Brushing properties	4321	3.4.3
Dipping properties	4341	3.4.3
Spraying properties	4331	3.4.3
Ash, percent by weight determination	5262	tables I and II
Color of resin (using 50 percent toluol and resin solution), Gardner	4248	table I
Directional reflectance	6121	3.4.4
Drying time	4061	3.4.5
Fineness of grind	4411	table III
Flexibility 1/	6221	3.4.11
Immersion resistance	6011	3.4.8
Nonvolatile and volatile content 2/	4041	tables I, II and III
Rosin, Lieberman Storch test	5031	table III
Self-lifting properties	6252	3.4.6
Specific gravity of resin	4183	table I
Specular gloss	6101	table III
Storage stability:		
Filled container 3/	3022	3.4.2.1
Partially filled container	3021	3.4.2.2
Consistency, Krebs-Stormer	4281	table III
Solvent analysis	7355, 7356	3.1

1/ Coating thickness 1.2 mils.
2/ Determination made at 105°C. for enamel and 293°C. for resin.
3/ Test first at 0°F. and repeat test at 105°F.

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4.4.2 Preparation of test panels.

4.4.2.1 Test panels. Unless otherwise specified herein, test panels shall be 3 inches wide and 5 inches long. Test panels shall be prepared in accordance with FED. TEST METHOD STD. No. 141, method 2011 solvent cleaned for steel panels, and method 2031 for wood panels. Wood panels shall be 5-ply, exterior-type, Douglas-fir plywood.

4.4.2.2 Coating of test panels. The test panels shall be pretreated and coated in accordance with MIL-T-704, type B. Unless otherwise specified herein, air drying of coats shall be at laboratory atmospheric conditions as specified in FED. TEST METHOD STD. No. 141, section 7. Each coat shall be allowed to air-dry 24 hours before subsequent coats are applied and 72 hours shall be allowed to elapse before testing of the coating system. The edges and backs of the system-coated metal panels shall be given one coat of spar varnish conforming to TT-V-121. The edges and backs of the system-coated wood panels shall be given two coats of spar varnish as specified above.

4.4.3 Accelerated weathering. Prepare two 3-inch X 9-inch, 20-gauge, 1020 cold rolled steel panels in accordance with 4.4.2.2, and then expose the panels for 300 hours to accelerated weathering in accordance with FED. TEST METHOD STD. No. 141, method 6151, or ASTM method G-26, type BH, and examine the paint film. Nonconformance to 3.4.10 shall constitute failure of this test.

4.4.4 Baking properties. A coated steel panel shall be baked for 30 minutes at plus 250°F. in an air-circulating oven. The baked film shall be examined for film irregularity. Evidence of film irregularity shall constitute failure of this test.

4.4.5 Softening point. The softening point of the resin shall be determined in accordance with ASTM Designation E 28. Nonconformance to the softening-point requirements specified in table I shall constitute failure of this test.

4.4.6 Salt spray resistance. This test will be performed in accordance with ASTM Designation B 117. Nonconformance to 3.4.9 shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging, packing and marking shall be examined to determine compliance with TT-P-143.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The enamel shall be packaged, packed and marked in accordance with TT-P-143. The level of packaging shall be level A or C and the level of packing shall be level A, B, or C as specified (see 6.2). The enamel shall be furnished in the quantity and size of container specified (see 6.2).

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5.1.1 Special marking. In addition to the marking specified in 5.1, eac' container shall be marked or labeled as may be required in accordance with the Manufacturing Chemists Association Manual L-1, Guide to the Precautionary Labeling of Hazardous Chemicals.

6. NOTES

6.1 Intended use. The enamel covered by this specification is intended for outside use where high humidity conditions exist as well as where partial or total immersion in fresh or salt water may occur. The enamel should be used over properly primed metal or wood surfaces.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2)
(c) Color required (see 3.4.4).
(d) Level of packaging and level of packing required (see 5.1).
(e) Size of container required (see 5.1).

6.3 The enamel covered by this specification should be purchased by volume, the unit being 1 U.S. liquid gallon of 231 cubic inches at 15.6°C. (60°F.).

MILITARY INTERESTS:

Custodians:

Army - ME
Air Force - 84

Review activity:

Army - MR

Civil Agency Coordinating Activities:

GSA - FSS

Preparing Activity:

Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

TT-E-522A Enamel, Phenolic, Outside

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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